

123456

A

B

C

D

123456

Power & PDN – rail summary

Rail	Nominal	Notes
VDD / VDDQ	1.2 V	DRAM core + I/O (VDDQ tied to VDD in v1)
VPP	2.5 V	DRAM wordline boost
VREFCA	0.6 V (VDD/2)	Command/address reference, external (VREFDQ is on-die on DDR4)
VDDSPD	(confirm)	SPD EEPROM supply
VSS / GND	0 V	Ground

Scope: this sheet covers power nets and references only –
no CA / data / SPD / connector signals here.
Decoupling/bypass and PDN (decap selection + placement, plane assignment)
are not enumerated in v1; PDN design is layout-stage work.

udimm-edge-interface

File: udimm-edge-interface.kicad_sch

spd-and-configuration

File: spd-and-configuration.kicad_sch

data-byte-lanes-dqs

File: data-byte-lanes-dqs.kicad_sch

address-command-clock

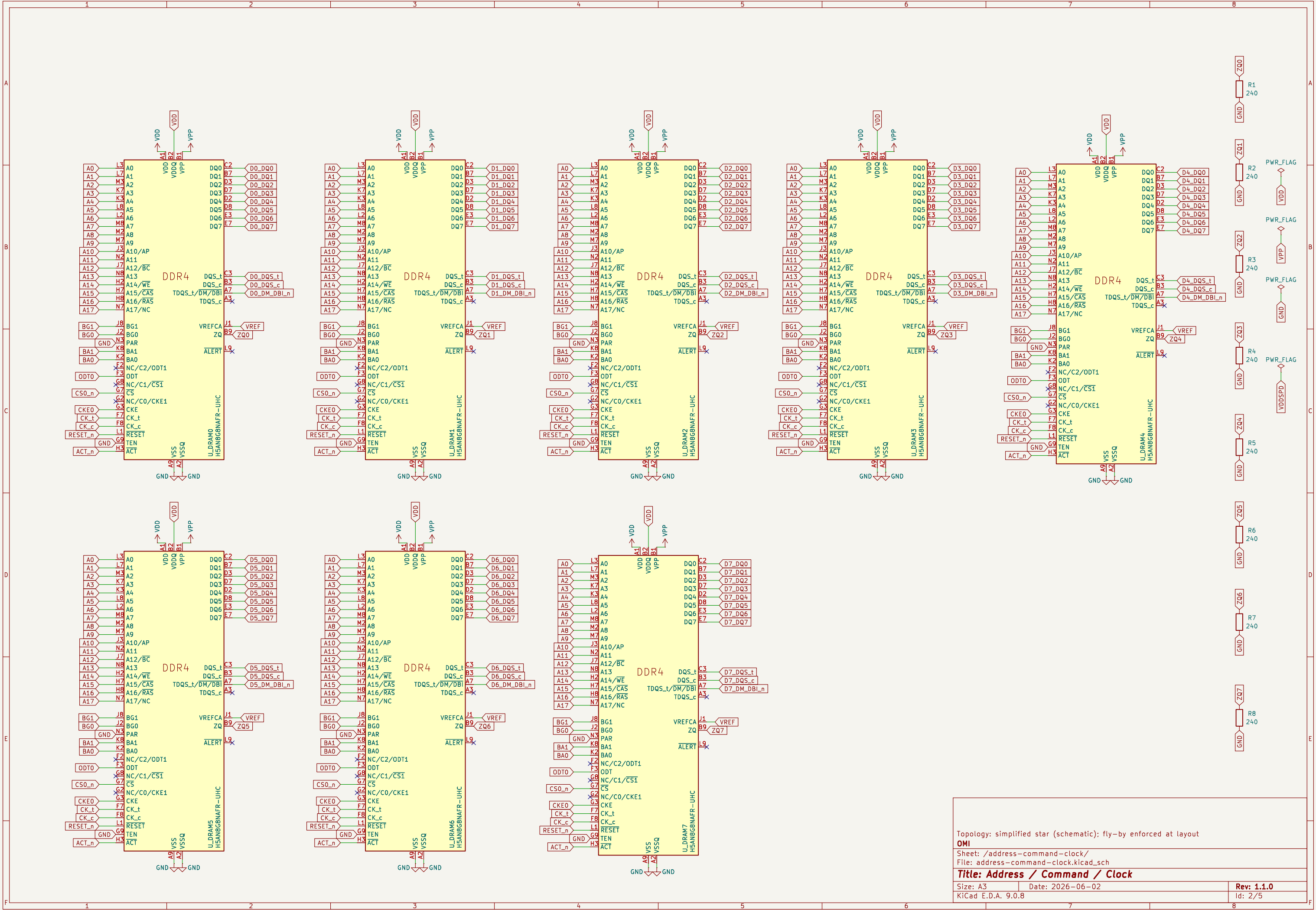
File: address-command-clock.kicad_sch

No CA, Data, SPD, or Connector signals allowed
Stage 7.1 – Power Schematic
OMI

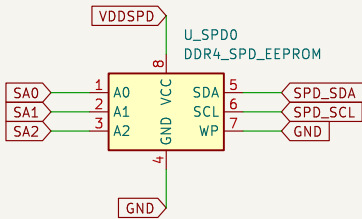
Sheet: /
File: omi_v1_power.kicad_sch

Title: Power & PDN

Size: A4	Date: 2026-06-02	Rev: 1.1.0
KiCad E.D.A. 9.0.8		Id: 1/5



Topology: simplified star (schematic); fly-by enforced at layout		
OMI		
Sheet: /address-command-clock/		
File: address-command-clock.kicad_sch		
Title: Address / Command / Clock		
Size: A3	Date: 2026-06-02	Rev: 1.1.0
KiCad E.D.A. 9.0.8		Id: 2/5



Assumptions:
- SMBus pull-ups for SPD_SCL and SPD_SDA are provided by host platform.
- WP is tied low (writes enabled) for OMI v1 development.
- SA0/SA1/SA2 are slot/address straps (host-side).

OMI

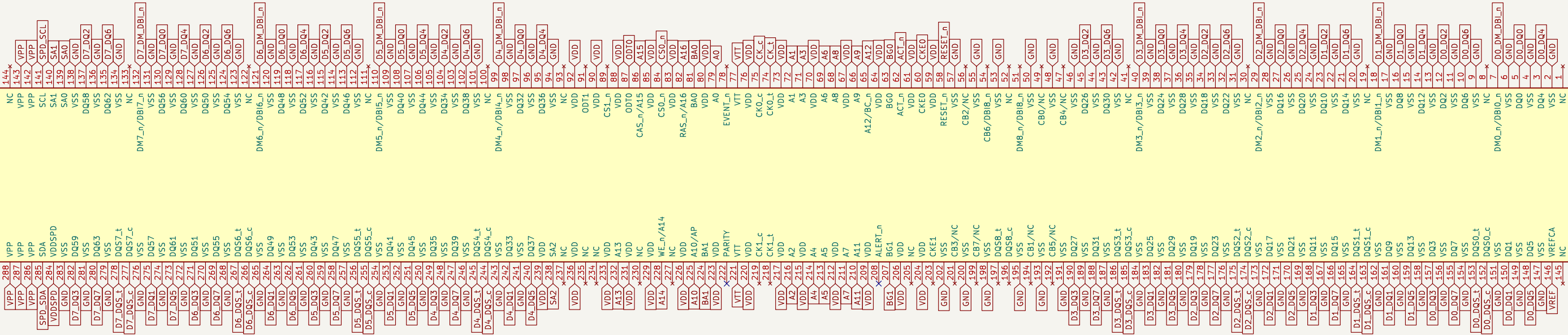
Sheet: /spd-and-configuration/
File: spd-and-configuration.kicad_sch

Title: SPD & Configuration

Size: A4	Date: 2026-06-02	Rev: 1.1.0
KiCad E.D.A. 9.0.8		Id: 4/5

UDIMM 288--Pin Edge Interface
J1 - DDR4 UDIMM 288-pin card edge.
Pin assignment per ddr4_udimm_288_pinmap.csv (keystone source of truth).
Mechanical outline, Edge.Cuts, and JEDEC key--notch position are NOT defined in v1.
Footprint is pads--only and NOT JEDEC--mechanically verified -- not fab--ready.

DDR4_UDIMM_288_Edge



J1